OTHM LEVEL 5 DIPLOMA IN LOGISTICS AND SUPPLY CHAIN MANAGEMENT

H/650/1106 - Procurement and Inventory Management

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SESSION 03

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OTHM LEVEL 5 DIPLOMA IN LOGISTICS AND SUPPLY CHAIN MANAGEMENT

SESSION 03

Learning Outcomes and Assessment Criteria

3. Understand the fundamentals of inventory management.	3.1 Differentiate between inventory management and inventory control. 3.2 Describe an inventory management cycle. 3.3 Compare different methods of inventory management. 3.4 Discuss the importance of inventory management to a business and its supply chain.

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3.1 Differentiate between inventory management and inventory control.

Inventory control vs. inventory management

	Inventory management	Inventory control
Scope	Broad scope	Well-defined scope
Focus	**Holistic activities** like planning and forecasting	**Targeted actions** like sending orders and receiving stocks
Goal	Meet market trends and improve profits	Achieve efficient stock management

Inventory management and inventory control are two distinct but closely related concepts within the aim of managing a company's inventory.

3.1 Differentiate between inventory management and inventory control.

Inventory Management: Inventory management refers to the overall process of overseeing and controlling the flow of materials, goods, and products into and out of a company's inventory. It involves planning, organizing, and overseeing all activities related to managing inventory levels.

Inventory Control: Inventory control, on the other hand, specifically focuses on the methods and procedures used to maintain optimal inventory levels within a company. It involves monitoring inventory levels, tracking stock movements, and implementing strategies to minimize excess inventory or stockouts.

3.1 Differentiate between inventory management and inventory control.

Inventory Management: Inventory management encompasses a broader scope of activities, including forecasting demand, ordering, receiving, storing, tracking, and managing inventory across the entire supply chain.

Inventory Control: Inventory control deals with more specific aspects of inventory management, such as setting reorder points, safety stock levels, and implementing inventory tracking systems.

Inventory Management: The primary objective of inventory management is to ensure that the right quantities of products are available at the right time, in the right place, and at the right cost to meet customer demand while minimizing inventory holding costs.
Inventory Control: Inventory control aims to optimize inventory levels by preventing stockouts, reducing excess inventory carrying costs, and improving overall operational efficiency.

3.1 Differentiate between inventory management and inventory control.

Inventory Management: Activities involved in inventory management include demand forecasting, procurement, storage, inventory tracking, order fulfillment, and inventory analysis.

Inventory Control: Inventory control activities include setting inventory policies, determining reorder points, establishing stock levels, implementing inventory management systems, conducting inventory audits, and optimizing inventory turnover.

3.1 Differentiate between inventory management and inventory control.

Inventory Management: Inventory management typically involves long-term planning and strategic decision-making to ensure the smooth flow of inventory over extended periods.

Inventory Control: Inventory control focuses more on short-term tactics and day-to-day operations to maintain optimal inventory levels in response to fluctuating demand and supply chain dynamics.

In summary, while inventory management encompasses the overall process of managing inventory, inventory control specifically deals with the methods and procedures used to maintain optimal inventory levels within a company's supply chain. Inventory control is a subset of inventory management, focusing on specific strategies and tactics to regulate inventory levels effectively.

3.1 Differentiate between inventory management and inventory control.

Let's consider an example of a retail clothing store to differentiate between inventory management and inventory control:

Inventory Management:

The clothing store needs to manage its inventory of various clothing items, including shirts, pants, and jackets.

Inventory Management Tasks:

Forecasting Demand: Using historical sales data and market trends to predict future demand for different clothing items.

Procurement: Ordering new inventory from suppliers based on demand forecasts and current inventory levels.

3.1 Differentiate between inventory management and inventory control.

Storage: Ensuring proper storage conditions to preserve the quality of clothing items and prevent damage.

Inventory Tracking: Implementing systems to track inventory levels, monitor stock movements, and identify slow-moving or obsolete items.

Order Fulfillment: Ensuring that customer orders are accurately fulfilled and delivered on time.

Inventory Analysis: Analyzing inventory data to identify patterns, optimize stocking levels, and make strategic decisions for the business.

3.1 Differentiate between inventory management and inventory control.

Inventory Control:

The clothing store aims to prevent stockouts of popular clothing items while minimizing excess inventory holding costs.

Inventory Control Tasks:

Setting Reorder Points: Determining the minimum inventory level at which new orders should be placed to avoid stockouts.

Safety Stock Management: Calculating and maintaining safety stock levels to buffer against unexpected fluctuations in demand or delays in supply.

Implementing Inventory Policies: Establishing guidelines and procedures for managing inventory, such as FIFO (First In, First Out) or JIT (Just-in-Time) inventory management.

3.1 Differentiate between inventory management and inventory control.

Inventory Optimization: Adjusting stocking levels and replenishment strategies based on sales patterns, seasonal trends, and supplier lead times to optimize inventory turnover and minimize carrying costs.

Inventory Audits: Conducting regular audits to verify inventory accuracy, identify discrepancies, and prevent shrinkage or theft.

Demand-Supply Alignment: Ensuring that inventory levels are aligned with customer demand and supplier capabilities to maintain a balanced supply chain.

In this example, inventory management involves the broader process of overseeing all aspects of the clothing store's inventory, including planning, procurement, storage, and analysis. Inventory control, on the other hand, focuses on specific strategies and tactics to regulate inventory levels effectively, such as setting reorder points, managing safety stock, and optimizing inventory turnover.

Types of inventories

1. **Raw Materials Inventory**

Basic materials used to produce finished goods.

Examples:
- Wood for furniture manufacturing
- Steel for car production
- Cotton for textile manufacturing

3.1 Differentiate between inventory management and inventory control.

Types of inventories

2. Work-In-Progress (WIP) Inventory

Items that are in the production process but are not yet completed.

- **Examples**:
  - A half-assembled car on an assembly line
  - Cut fabric pieces in a garment factory
  - Circuit boards being soldered in an electronics factory

Types of inventories

3. Finished Goods Inventory

Products that are completed and ready for sale.

- **Examples**:
  - Televisions ready for shipment in a warehouse
  - Bottled beverages in a distributor's storage
  - Packaged shoes on a retail shelf

Types of inventories

4. Maintenance, Repair, and Operations (MRO) Inventory

Supplies used to support the production process but are not part of the finished product.

- **Examples**:
  - Lubricants, cleaning supplies, gloves
  - Spare parts for machines
  - Tools used in a factory

Types of inventories

5. Transit Inventory (Pipeline Inventory)

Inventory that is currently being transported from one location to another.

- **Examples**:
  - Electronics being shipped from China to the U.S.
  - Crude oil on a tanker en route to a refinery
  - Books being delivered from a warehouse to a retailer

Types of inventories

6. Buffer Inventory (Safety Stock)

Extra inventory kept on hand to guard against uncertainties in demand or supply.

- **Examples**:
  - Additional food supplies before a festive season
  - Extra pharmaceuticals in case of delivery delays
  - Backup stock of parts in a car manufacturing plant

3.1 Differentiate between inventory management and inventory control.

Types of inventories

7. Anticipation Inventory

Inventory built up in expectation of future demand.

- **Examples**:
  - Toys manufactured in advance of the holiday season
  - Seasonal clothing stocked before winter
  - School supplies stocked before back-to-school season

Types of inventories

8. Decoupling Inventory

Inventory stored between different stages of production to ensure smooth operations.

- **Examples**:
  - A buffer of semi-finished parts between two machines in a production line
  - Car frames stocked before assembly of engines

Inventory control is a warehousing function

Inventory control is a core warehousing function that involves **managing and regulating the flow of goods into, within, and out of a warehouse** to ensure the right quantity of the right items is available at the right time, while minimizing costs and preventing losses.

It focuses on:

- Maintaining optimal stock levels
- Preventing overstocking or stockouts
- Ensuring accurate tracking and documentation
- Improving efficiency and reducing waste

Inventory control is a warehousing function

1. **Stock Level Management**

Purpose: To maintain the right amount of inventory to meet customer demand without overstocking.

Example:

- A warehouse for a smartphone company keeps 10,000 units of a popular model based on historical sales data.
- Inventory control monitors sales trends and alerts managers when stock falls below 2,000 units (reorder point), prompting a restock.

3.1 Differentiate between inventory management and inventory control.

Inventory control is a warehousing function

2. Stock Rotation and Movement

Purpose: Ensures that older stock is used/sold first to avoid spoilage or obsolescence, often through methods like FIFO (First-In, First-Out) or LIFO (Last-In, First-Out).

Example:

- In a food warehouse, dairy products are rotated using FIFO so that older milk cartons are sold first, reducing spoilage.

Inventory control is a warehousing function

3. Inventory Accuracy and Record-Keeping

Purpose: To ensure that the physical stock matches the inventory records (manual or digital), which is crucial for order fulfillment and financial reporting.

Example:
- A warehouse uses barcode scanners and inventory management software to update records every time items are received or dispatched.
- Regular cycle counts (e.g., weekly) are done to verify physical counts against records.

Inventory control is a warehousing function

4. Reorder Point Determination

Purpose: Helps determine the ideal stock level at which a new order should be placed to avoid stockouts.

Example:

- An auto parts warehouse sets a reorder point of 500 brake pads. Once stock reaches this level, the system automatically alerts procurement to place a new order based on lead time and usage rate.

Inventory control is a warehousing function

5. Minimizing Holding Costs

Purpose: Reduces the costs associated with storing unsold goods, such as rent, insurance, depreciation, and obsolescence.

Example:
- A fashion retail warehouse avoids over-ordering trendy items that may go out of style quickly, minimizing the risk of unsold inventory that ties up space and capital.

3.1 Differentiate between inventory management and inventory control.

Inventory control is a warehousing function

6. Loss Prevention and Security

Purpose: Prevents theft, damage, or misplacement of goods within the warehouse.
Example:

- A warehouse installs CCTV cameras and restricts access to high-value electronics storage areas. Inventory audits are conducted monthly to detect any discrepancies.

3.1 Differentiate between inventory management and inventory control.

Inventory control is a warehousing function

7. Classification of Inventory (ABC Analysis)

Purpose: Focuses control efforts based on the importance and value of items.
Example:

- In a pharmaceutical warehouse:
  - "A" items (e.g., life-saving drugs) are closely monitored and frequently restocked.
  - "B" items (less critical, moderate demand) are monitored monthly.
  - "C" items (low cost, low demand) are checked quarterly.

3.1 Differentiate between inventory management and inventory control.

Inventory control is a warehousing function

8. Supporting Order Fulfillment

Purpose: Ensures that orders can be picked, packed, and shipped efficiently and accurately.

Example:
- A warehouse serving an e-commerce store uses a well-organized layout and real-time inventory tracking so that customer orders are picked quickly and shipped same-day.

3.1 Differentiate between inventory management and inventory control.

Inventory control is a warehousing function

Benefits of Effective Inventory Control in Warehousing

- Reduces operational costs
- Improves customer satisfaction through timely order fulfillment
- Prevents stockouts and overstocking
- Enhances warehouse space utilization
- Enables better forecasting and decision-making

3.2 Describe an inventory management cycle

An inventory management cycle refers to the continuous process of overseeing and controlling the flow of goods or materials within an organization's inventory system. This cycle typically involves several interconnected stages, each contributing to the efficient management of inventory.

Demand Forecasting:

The cycle often begins with demand forecasting, where historical sales data, market trends, and other factors are analyzed to predict future demand for products.

Demand forecasting helps in planning inventory levels and ensuring that sufficient stock is available to meet customer demand without overstocking.

3.2 Describe an inventory management cycle

Inventory Planning:

Based on demand forecasts and sales projections, inventory planning involves setting inventory targets, determining stocking levels, and establishing inventory policies.

Inventory planning considers factors such as lead times, supplier capabilities, storage capacity, and budget constraints to optimize inventory levels.

Procurement:

Once inventory needs are determined, the procurement stage involves ordering or acquiring goods from suppliers to replenish inventory levels.

Procurement activities may include negotiating with suppliers, placing purchase orders, scheduling deliveries, and coordinating with logistics providers.

3.2 Describe an inventory management cycle

Receiving and Inspection:

Upon receipt of goods from suppliers, the inventory is inspected to ensure that the delivered items match the order specifications and meet quality standards.

Receiving and inspection activities involve verifying quantities, checking for damages or defects, and recording receipt details in the inventory management system.

3.2 Describe an inventory management cycle

Storage and Organization:

After inspection, the received inventory is properly stored and organized within the warehouse or storage facility.

Storage and organization activities include assigning storage locations, labeling items for easy identification, and optimizing warehouse layout for efficient picking and replenishment.

3.2 Describe an inventory management cycle

Inventory Tracking and Control:

Throughout the inventory management cycle, inventory levels are continuously monitored and tracked to ensure accuracy and control.

Inventory tracking involves using barcode systems, RFID technology ‘Radio Frequency Identification (RFID) refers to a wireless system comprised of two components: tags and readers’, or inventory management software to record stock movements, track inventory locations, and update inventory records in real-time.

3.2 Describe an inventory management cycle

Order Fulfillment:

When customer orders are received, the inventory management system generates pick lists or work orders to fulfill the orders from available stock.

Order fulfillment activities involve picking, packing, and shipping products to customers in a timely and accurate manner.

Inventory Analysis and Optimization:

Periodic inventory analysis is conducted to assess inventory performance, identify trends, and optimize inventory levels.

Inventory optimization strategies may involve adjusting reorder points, minimizing excess inventory, liquidating slow-moving items, or implementing inventory turnover improvement initiatives.

Cycle Counting and Auditing:

To maintain inventory accuracy, regular cycle counting and auditing are performed to reconcile physical inventory counts with inventory records.

Cycle counting involves counting a subset of inventory items on a rotating basis, while audits involve comprehensive reviews of inventory records and processes.

Continuous Improvement:

The inventory management cycle is a continuous process that requires ongoing evaluation and improvement.

Feedback from various stages of the cycle, along with performance metrics and key performance indicators (KPIs), is used to identify areas for improvement and implement corrective actions to enhance efficiency and effectiveness.

By following this inventory management cycle, organizations can ensure the smooth flow of goods or materials throughout the supply chain, minimize stockouts and excess inventory, optimize inventory investment, and improve overall operational performance.

3.3 Compare different methods of inventory management.

Various methods of inventory management exist, each with its own advantages, disadvantages, and suitability for different types of businesses and industries

Just-in-Time (JIT) Inventory Management:

JIT inventory management aims to minimize inventory levels by synchronizing production with customer demand. It involves receiving goods only as they are needed in the production process, reducing storage costs and inventory holding costs.

Advantages:

Reduces holding costs associated with excess inventory.

Improves cash flow by minimizing tied-up capital.

Encourages leaner and more efficient production processes.

Disadvantages:

Vulnerable to supply chain disruptions, as any delays in the supply chain can lead to stockouts.

Requires highly reliable suppliers and efficient logistics to ensure timely delivery of materials.

Limited applicability in industries with high demand variability or long lead times.

3.3 Compare different methods of inventory management.

ABC Analysis:

Description: ABC analysis categorizes inventory items into three categories based on their value and contribution to overall inventory costs: A (high-value items), B (moderate-value items), and C (low-value items). Different inventory management strategies are then applied to each category based on their importance.

Advantages:

Helps prioritize inventory management efforts by focusing on high-value items.

Provides a systematic approach to allocating resources and setting inventory policies.

Facilitates better inventory control and cost management.

Disadvantages:

Requires accurate and up-to-date data on item values and usage patterns.

May overlook non-monetary factors such as lead times or criticality.

Can be time-consuming to maintain and update classifications regularly.

3.3 Compare different methods of inventory management.

Economic Order Quantity (EOQ):

EOQ is a mathematical model used to determine the optimal order quantity that minimizes total inventory costs, including ordering costs and holding costs. It calculates the ideal balance between ordering larger quantities to benefit from economies of scale and minimizing holding costs associated with excess inventory.

Advantages:

Helps minimize total inventory costs by determining the most cost-effective order quantity.

Provides a straightforward formula for calculating optimal inventory levels.

Suitable for businesses with stable demand and known ordering costs.

Disadvantages:

Assumes static demand and stable lead times, which may not reflect real-world variability.

Ignores factors such as stockouts, storage constraints, or supplier constraints.

Requires accurate data on ordering costs, holding costs, and demand patterns.

3.3 Compare different methods of inventory management.

Vendor Managed Inventory (VMI):

In VMI, the supplier takes responsibility for managing the customer's inventory levels. The supplier monitors inventory levels at the customer's site and automatically replenishes stock as needed, based on pre-established agreements and demand forecasts.

Advantages:

Reduces inventory holding costs and stockouts for the customer.

Improves supply chain visibility and collaboration between suppliers and customers.

Shifts inventory management responsibilities to the supplier, freeing up resources for the customer.

Disadvantages:

Requires a high level of trust and collaboration between the supplier and the customer.

May lead to increased dependency on the supplier and reduced flexibility in sourcing.

Requires robust communication channels and reliable technology for data exchange.

Cross-Docking:

Description: Cross-docking involves transferring goods directly from inbound to outbound transportation vehicles with minimal or no storage in between. It aims to reduce handling and storage costs by streamlining the flow of goods through the supply chain.

Advantages:

Reduces storage costs and inventory holding costs.

Improves order fulfillment speed and reduces lead times.

Minimizes the risk of inventory obsolescence or damage.

3.3 Compare different methods of inventory management.

Disadvantages:

Requires sophisticated logistics and coordination between suppliers, carriers, and distribution centers.

May be less suitable for businesses with complex or varied product assortments.

Vulnerable to disruptions in transportation or supply chain logistics.

These methods represent just a few of the many approaches to inventory management, each offering unique benefits and challenges depending on the specific needs and circumstances of a business. Effective inventory management often involves a combination of these methods tailored to the organization's goals, industry dynamics, and operational requirements.

3.3 Compare different methods of inventory management.

Drop shipping is a retail fulfillment method where a **store doesn't keep the products it sells in stock**. Instead, when a customer places an order, the store **purchases the item from a third party** (usually a wholesaler or manufacturer), who then **ships it directly to the customer**.

How Drop Shipping Works (Step-by-Step)

Customer places an order on the seller’s online store.

Seller forwards the order to a supplier or manufacturer.

Supplier ships the product directly to the customer.

Seller earns profit by charging the customer more than the supplier charges them.

Example

Imagine an online store called **TrendyT-Shirts.com** that sells graphic t-shirts.

A customer orders a t-shirt for **$25** on TrendyT-Shirts.com.

TrendyT-Shirts doesn't keep any shirts in stock. Instead, it forwards the order to a supplier who makes the shirt for **$15**.

The supplier **prints the shirt and ships it directly** to the customer.

TrendyT-Shirts **keeps a $10 profit** ($25 sale price – $15 supplier cost).
The seller focuses on marketing, branding, and customer service — while the supplier handles production and logistics.

3.4 Discuss the importance of inventory management to a business and its supply chain

Inventory management plays a crucial role in the success and efficiency of both businesses and their supply chains. Here are several reasons why inventory management is important

Optimizing Working Capital:

Effective inventory management helps optimize working capital by ensuring that the right amount of capital is invested in inventory at any given time. By minimizing excess inventory and stockouts, businesses can allocate their resources more efficiently, freeing up capital for other investments or operational needs.

Meeting Customer Demand:

Maintaining optimal inventory levels enables businesses to meet customer demand promptly and efficiently. By having the right products available when customers need them, businesses can enhance customer satisfaction, build loyalty, and capture market share.

Reducing Holding Costs:

Inventory carrying costs, including storage, insurance, and obsolescence costs, can represent a significant financial burden for businesses. Effective inventory management helps minimize these holding costs by optimizing inventory levels, reducing excess stock, and improving inventory turnover rates.

Minimizing Stockouts and Backorders:

Stockouts and backorders can result in lost sales, dissatisfied customers, and damage to a company's reputation. Inventory management strategies such as safety stock management and demand forecasting help minimize the risk of stockouts, ensuring that products are available when customers need them.

3.4 Discuss the importance of inventory management to a business and its supply chain

Improving Supply Chain Efficiency:

Inventory management is essential for optimizing supply chain efficiency and performance. By streamlining inventory flows, coordinating production schedules, and aligning inventory levels with demand forecasts, businesses can reduce lead times, minimize waste, and improve overall supply chain responsiveness.

Enhancing Production Planning and Control:

Effective inventory management provides valuable insights into demand patterns, production requirements, and resource allocation. By aligning inventory levels with production schedules and capacity constraints, businesses can optimize production planning and control, minimize production disruptions, and improve resource utilization.

Mitigating Risks and Uncertainties:

Inventory management helps mitigate risks and uncertainties associated with supply chain disruptions, demand volatility, and market fluctuations. By implementing robust inventory control measures, businesses can build resilience, adapt to changing market conditions, and minimize the impact of unforeseen events on their operations.

Supporting Strategic Decision-Making:

Inventory management data provides valuable inputs for strategic decision-making, such as product portfolio management, pricing strategies, and expansion plans. By analyzing inventory trends, demand patterns, and performance metrics, businesses can make informed decisions to optimize their operations and drive long-term growth.

In summary, effective inventory management is essential for businesses to maintain competitiveness, improve operational efficiency, and maximize profitability. By optimizing inventory levels, minimizing holding costs, and enhancing supply chain coordination, businesses can meet customer demand, reduce risks, and achieve sustainable growth in today's dynamic and competitive business environment.

3.4 Discuss the importance of inventory management to a business and its supply chain

Example:

Consider a retail clothing chain with multiple stores across a region. Effective inventory management is critical to its success.

Demand Forecasting: By analyzing historical sales data, seasonal trends, and market conditions, the retailer can accurately forecast demand for different clothing items. This allows them to stock the right quantities of each product in each store, minimizing stockouts and excess inventory.

Supplier Relationships: The retailer maintains strong relationships with suppliers to ensure timely delivery of goods. They may use techniques like vendor-managed inventory (VMI) or just-in-time (JIT) inventory to optimize supply chain efficiency and reduce lead times.

3.4 Discuss the importance of inventory management to a business and its supply chain

Inventory Optimization: The retailer regularly monitors inventory levels in each store and adjusts replenishment orders accordingly. They use inventory management software to track stock levels, sales data, and reorder points, enabling them to maintain optimal inventory levels while minimizing holding costs.

Seasonal Variations: During peak seasons like holidays or back-to-school shopping, demand for certain clothing items may increase significantly. The retailer adjusts their inventory levels and procurement strategies to meet this increased demand without overstocking.

Promotions and Discounts: Inventory management also plays a crucial role in planning promotions and discounts. By analyzing inventory levels and sales data, the retailer can identify slow-moving items and offer targeted promotions to clear excess stock and maximize sales.

3.4 Discuss the importance of inventory management to a business and its supply chain

In this example, effective inventory management enables the retail clothing chain to optimize cash flow, minimize costs, improve customer service, and make informed business decisions, ultimately contributing to its overall success